



# Multi-mode vortex-induced vibration of a long-span bridge under non-uniform flows

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## ABSTRACT

An increasing number of modern long-span bridges have been built in complex valley terrain, and the influence of non-uniform flows on the vortex-induced vibrations (VIVs) of bridges should be considered. Therefore, this study investigated the multi-mode VIVs of long-span bridges in both the completed bridge stage and erection stage using a taut-strip model under a uniform flow and two types of non-uniform flow. The non-uniform flows were generated by adjusting spires (both their separation distance and shapes) in wind tunnel. The two non-uniform flows were represented by the linear and parabolic wind velocity flow distributions. The characteristics of the vibration responses and wake flows, mechanism of multi-mode VIVs, and aerodynamic differences between different stages and under different incoming flows were discussed. The results indicated that the occurrence of multi-mode VIVs arose from a wake frequency difference covering two or more modal frequencies owing to variations in the wind profile. The non-uniform flows disturb the regular vortices in the wake due to the variable wind velocity and higher turbulence intensity, leading to suppression of the VIV response.

## 1. Introduction

With the increasing length of long-span bridges in recent years, the main spans become longer, more flexible, sensitive, and prone to exhibit oscillations at low wind speeds [16,17]. Vortex-induced vibration (VIV) occurs when the vortex-shedding frequency matches one of the natural frequencies of flexible structures such as marine risers [26,21], stay cables [6–8,22], and modern long-span bridges [16,11,17,37]. The VIV is a self-excited and self-limited vibration, and one of its classical characteristics is the lock-in phenomenon [4,26,29]. The lock-in phenomenon has been investigated by many scholars, including Bearman [4] and Williamson and Govardhan [30]. The maximum vibration amplitudes occur within the lock-in region under a range of reduced velocities [4]. The frequent vibration amplitudes of the bridge may cause discomfort to drivers, even lead to structural fatigue [16,9,37]. Most of the existing researches focus on evaluating the VIV of flexible bridges [34,35,37] under steady and uniform incoming conditions, while the dynamic characteristics under non-uniform flow still need further understanding.

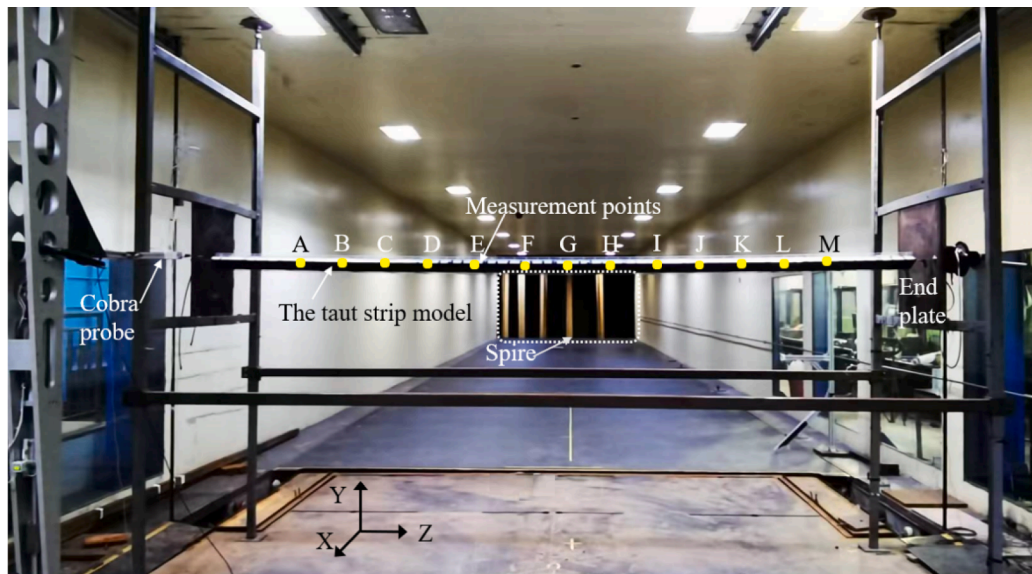
Owing to the flexibility and serried modes of long-span bridges, vertical VIVs of long-span bridges can be excited mode-by-mode when the wind speed increases gradually [37]. The incoming flow and turbulence intensity along the spanwise are not uniform in engineering [17,18], and the existing VIV response investigations seldom consider the nonuniformity of the incoming flow. However,

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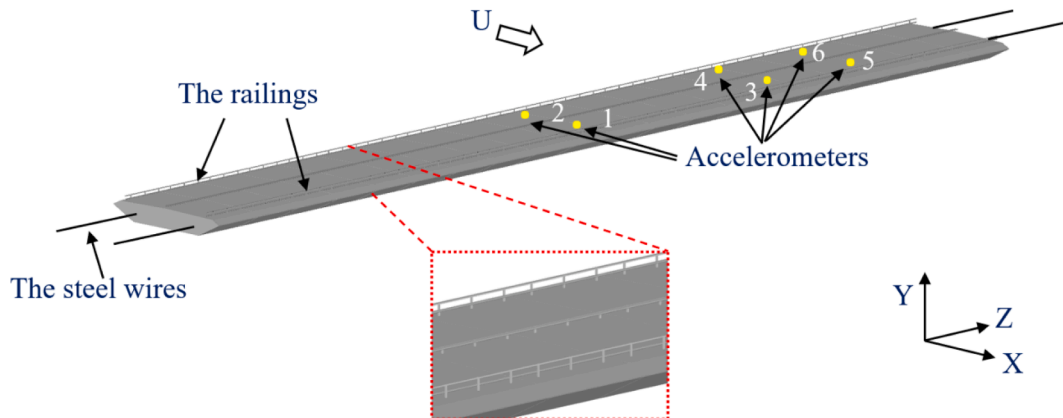
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based on the full-scale measurements, the surrounding wind field is complex and inhomogeneous [18] due to the variety of deep ravine terrain. To further understand the wind characteristics around bridges across deep ravines, the difference between the results from monitoring systems over two years and experiments was investigated by Bastos et al. [3]. The comparison considered the mean wind directions and velocities, turbulence intensities, and their spanwise components, and the results showed that the turbulence intensities were non-uniformly distributed along the span owing to the complex terrain. Lystad et al. [23] investigated wind field spatial variations for the spanwise mean wind velocity and turbulence intensity along the Hardanger Bridge girder, the comparison between the full-scale measurements and the terrain model wind tunnel tests both show that the spanwise wind profiles and along-wind turbulence intensity are non-uniform. A linear and a parabolic trend wind profile were observed in different inflow directions through the full-scale measurements. Zhang et al. [36] identified the dynamic properties of the bridge and the characteristics of the VIV event by analyzing monitoring data, and the phenomenon of the dominant vibration mode switch was observed. The 3rd and 4th structural modes alternatively appear in the time–frequency spectrum for a short period.

Many investigations concerning the multi-mode VIVs of flexible structures have been conducted in recent decades. Chen et al. [8] conducted an experiment on the VIVs of a flexible cable model under different wind speed profiles. The results indicated that multi-mode VIVs occurred because the difference in the wake frequencies caused by the velocity profile could cover two or more modal frequencies. When multi-mode VIVs occur, the modal frequencies alternately appear under non-uniform flow owing to the mutual interference among the frequencies. Chen et al. [10] identified and analyzed the multi-mode VIVs of slender risers using an empirical mode decomposition (EMD) method and an experiment on a large-scale flexible pipe. When the reduced velocity was sufficiently high,



(a)



(b)

**Fig. 1.** Experiment setup: (a) Experiment setup of the taut-strip model (with railings) and locations of 13 measurement points (A–M); (b) Schematic of the taut-strip model (with railings) and locations of six accelerometers (1–6).

the phenomena of energy transfer, mode jumping, and multiple modes appeared simultaneously. The 1st through 13th modes were observed, and multiple dominant modes of the VIVs occurred both in the IL and CF directions. Zhu et al. [38] conducted an experiment on a curved flexible cylinder to investigate the VIVs under a log-low shear flow, and the results indicated that the dominant modal frequencies of the cylinder varied along the length, while the competition between the multiple modes interrupted the lock-in phenomenon. Based on the numerical approach, Jiang et al. [14] investigated the multi-mode response of the VIVs of risers in shear cross flows, and the results showed that some of the structural frequencies of the risers were locked-in at different locations along the length.

For the bridge structures, researchers normally studied the single-mode VIVs under uniform and turbulent flows around long-span bridges. Fujino et al. [12] analyzed a full-scale ten-span bridge model in uniform and turbulent flows and observed that the 1st to 9th modes of the VIVs were present not only in the uniform flow but also in two different turbulent flows. The maximum vibration amplitudes of a small portion of the VIV modes under uniform flows were lower than those under turbulent flows. Zhu et al. [39] put forward a mode-by-mode VIV analysis method and analyzed the VIV response of a section model under uniform mean wind as well as non-uniformly distributed mean and turbulent winds individually. Comparing the differences between uniform and turbulent winds, the results indicated that the mean wind speed with non-uniformly distribution diminished the vertical VIV responses.

The investigations paid little attention to the studies on the multi-mode VIVs of the long-span bridges. When multi-mode VIVs of other flexible structures occur, these several modes compete with each other resulting in more complicated frequency components. Higher frequency components in lower modal VIVs may cause fatigue damage accumulation in the structure, which should be avoided [19]. Therefore, it is a challenge to predict the frequency patterns and other characteristics during VIV processes [2]. For bridge structures, the characteristics, mechanism, and identification method of the multi-mode VIVs need further investigation.

In this paper, the characteristics of the vibration responses and wake flows and aerodynamic differences between different stages and under different incoming flows are discussed, mechanism and identification method of multi-mode VIVs are given. The remainder of this paper is organized as follows: in Section 2, the experimental setup of the taut-strip model and the designs of two non-uniform flows are introduced. In Section 3, the vibration and wake characteristics of the VIV responses of the taut-strip model under uniform and non-uniform flows are discussed and a prediction approach of the wake frequencies and an identification method of the multi-mode VIV of the flexible bridge without railings are proposed. Finally, conclusions are drawn in Section 4.

## 2. Experimental setup

Wind tunnel experiments were performed at the Joint Laboratory of Wind Tunnel and Wave Flume (WTWF) at the Harbin Institute of Technology, P.R. China. The cross-section size of the test section is 3.6 m (height)  $\times$  6.0 m (width), with a length of 50.0 m. The maximum wind speed is 30 m/s. In this study, the tested wind speed varied from 0.8 to 12 m/s at velocity intervals of 0.1 m/s, the wind speed range is suitable enough to investigate the characteristic of VIV responses.

### 2.1. Taut-strip model

Due to the space limitation and response request, a taut-strip model of the Storebælt suspension bridge with a scale ratio of 1:60 was used in the wind tunnel tests, as shown in Fig. 1. As shown in Fig. 1(a), the origin of the Cartesian coordinate system is located at the left end of the taut-strip model, the x-axis is the positive streamwise direction, the y-axis is perpendicular to the incoming flow, and the z-axis corresponds to the spanwise direction along the bridge. The length of the taut-strip model was 4000 mm, the aspect ratio of the whole model was 7.74, and the spanwise distance between each section model was 1 mm. The model mass was 3.65 kg/m, and the blockage ratio was 1.3 %. Two end plates were used to prevent the model from being influenced by the disturbed flow. Two steel wires with a diameter of 8 mm were passed through the taut-strip model to provide structural stiffness. The distance between the steel wires was 315 mm, and the tension of the two steel wires was identical. Two horizontal steel beams are used to make sure the stability of the frame, the vertical distance between the steel beams and the taut-strip model is high enough, therefore, the flow fields around the model will not be affected. The taut-strip model was composed of 20 sections, as shown in Fig. 1(a) and (b). Each section model was made of polystyrene foam plates covered with acrylonitrile butadiene styrene plastic plates to provide a lightweight section model with shape invariance. At a scale of 1:60, the width, height, and length of each section model were 516.67 mm, 72.33 mm, and 199 mm,

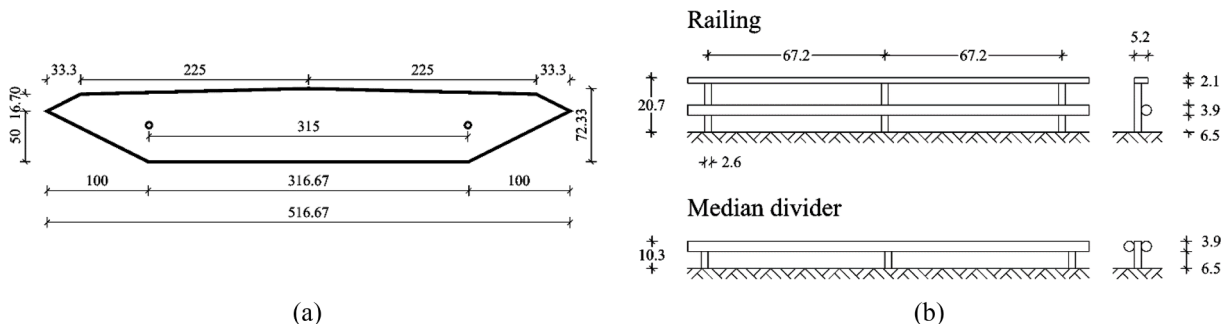


Fig. 2. Parameter settings: (a) Geometric Parameters of the model; (b) Parameters of the railing and median divider (unit: mm).

respectively, as shown in Fig. 2(a).

The aerodynamic characteristics and VIVs of a bridge are very sensitive to its geometrical variation [5,13], these bridge attachments such as the railings and median dividers may affect the responses on the bridge surface, the VIVs of the bridge with attachments are quite different compared with a bare bridge [15]. According to Larsen et al. [16], a long-span bridge can experience VIVs during not only the completed bridge stage but also the erection stage. Therefore, it is of great significance to investigate the multi-mode VIVs with and without railings and median dividers, railings and a median divider were installed on the upper surface along the span, as shown in Fig. 2(b), referring to Larsen et al. [16]. The mode shapes of the bridge model are plotted in Fig. 3. The geometric dimensions and system parameters of the taut-strip model are listed in Table 1, where the first and second structural modal damping ratios were obtained through free vibration tests. Due to the rapidly decaying signal, the third modal damping ratio is difficult to be obtained through free vibration tests. Therefore, the third modal damping ratio was calculated using the Rayleigh proportion damping model,  $[C] = \alpha[M] + \beta[K]$ .

## 2.2. Measurement equipment

Six accelerometers (Brüel and Kjær 4507B) were installed at positions 1/2, 1/4, and 1/6 of the total length of the model. Two accelerometers were employed to measure the vibration responses of the taut-strip model and were symmetrically arranged at each position, as illustrated in Fig. 1(b). The positions at 1/2, 1/4, and 1/6 of the model length measured the first to third modal structural vibration responses, respectively. The sampling frequency and time were 1000 Hz and 60 s, respectively.

A cobra probe (TFI Pty Ltd, North-East Victoria, Australia) was used to measure the incoming wind speed, and the distance between the cobra probe and the leading edge of the taut-strip model was 1500 mm. Thirteen measurement points are shown in Fig. 1(a), denoted A to M; these were used to obtain the near-wake velocity data along the span, and the distance between two adjacent points was 250 mm. The sampling frequency and time were 2500 Hz and 40 s, respectively.

To describe the two non-uniform wind speed profiles conveniently, the wind speed range that varied from Point A to M was used to represent the linear wind speed distribution range according to the shape of the wind velocity profiles. The wind velocity range that varied from Point A to G was used to describe the parabolic wind speed distribution range owing to its symmetry.

## 2.3. Non-uniform wind velocity profile and cases

To satisfy the growing demand for relieving traffic pressure, an increasing number of long-span bridges have been built in coastal and complex valley areas. According to the full-scale measurements [23], the linear and parabolic wind velocity profiles are both observed along the span in different wind directions. Therefore, two non-uniform wind velocities were designed to correspond to the different wind directions and complex terrains. The incoming flows were simplified as linear and symmetrical parabolic wind velocity profiles, respectively, as shown in Fig. 4. To obtain these wind velocity profiles, arrays of rectangular spires were arranged with different spacings in the wind tunnel. After a series of attempts, five and six spires were used to simulate the linear and parabolic velocity profiles, respectively, as shown in Fig. 4.

The velocity and turbulence intensity profiles of two non-uniform flows are shown in Fig. 5. Due to the variable velocities along the span, the wind speed at the midspan position was selected as the reference point to describe the non-uniform incoming flow for convenience. It is more susceptible for the middle spanwise location of the bridge to be affected by winds [27], and many field measurements also pay more attention to the data at the midspan position [18,23]. Three linear wind velocity profiles obtained by changing the wind velocities of the incoming flow are shown in Fig. 5(a). It can be seen that the flow velocities increase linearly along

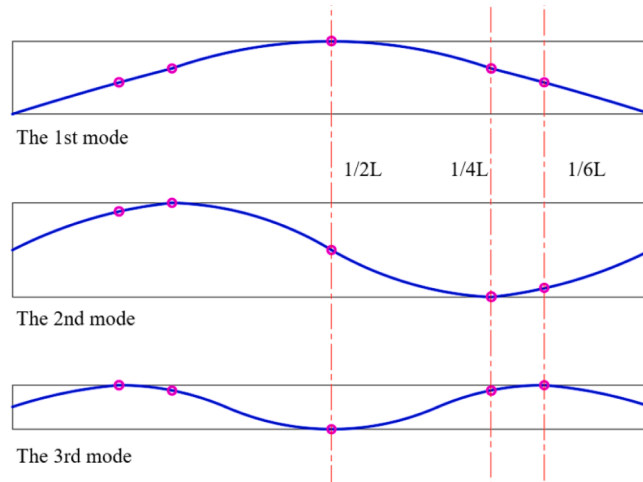
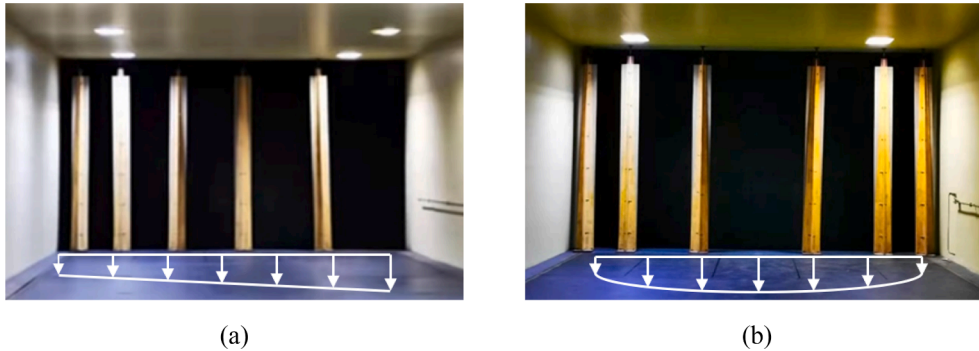


Fig. 3. Mode shapes of the bridge model.



**Table 1**  
Parameters of the test model.

| Parameter     |          | Symbol  | Unit | Value |
|---------------|----------|---------|------|-------|
| Length        |          | $L$     | mm   | 4000  |
| Width         |          | $D$     | mm   | 516.7 |
| Height        |          | $H$     | mm   | 72.3  |
| mass          |          | $m$     | Kg/m | 3.65  |
| Frequency     | 1st mode | $f_1$   | Hz   | 3.4   |
|               | 2nd mode | $f_2$   | Hz   | 6.8   |
|               | 3rd mode | $f_3$   | Hz   | 10.2  |
| Damping ratio | 1st mode | $\xi_1$ | %    | 0.33  |
|               | 2nd mode | $\xi_2$ | %    | 0.17  |
|               | 3rd mode | $\xi_3$ | %    | 1.2   |



**Fig. 4.** Design of the wind velocity distributions and arrangements of spires: (a) linear and (b) parabolic wind velocity distributions.

the span. Referring to Fig. 4(a), the trends of the three linear wind velocity profiles are consistent with the trend of the designed profile. The three different turbulence intensity profiles almost coincide with each other, and the trends are opposite to those of the wind velocity profiles as shown in Fig. 5(c). Similarly, the symmetrical parabolic wind velocity profiles were obtained, and the corresponding turbulence intensity profiles were almost unchanged, as shown in Fig. 5(b) and (d). In addition, the parabolic trend of the wind velocity profile became more obvious with an increase in the wind speed.

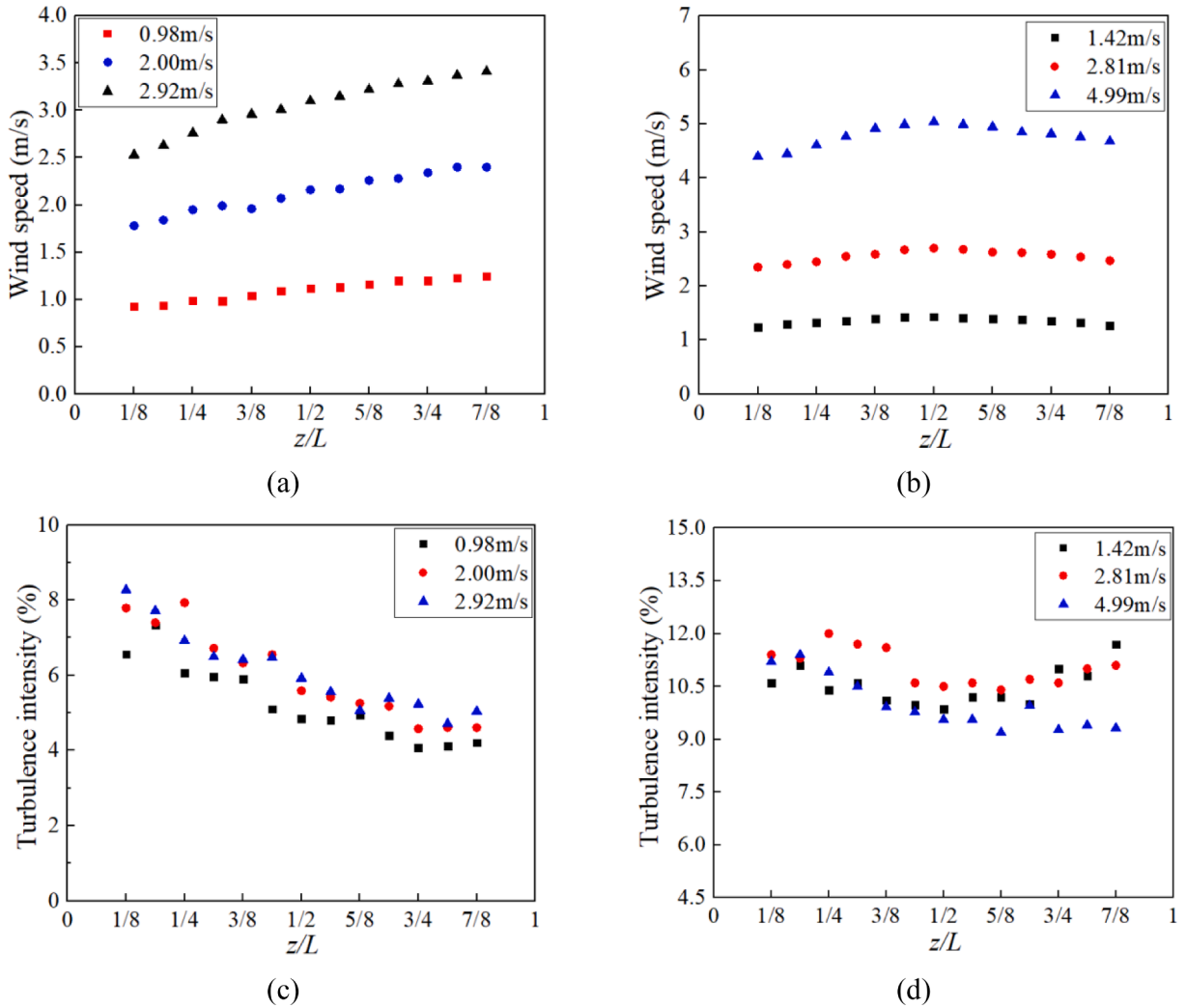
The average streamwise ( $I_u$ ) and vertical ( $I_w$ ) turbulence intensities vary with wind velocities at the midspan position are shown in Fig. 6, the turbulence intensities at the other measurement points can be roughly estimated through the trend in Fig. 5(c) and (d). The  $I_u$  and  $I_w$  of the uniform flow stabilize at 1.9 % and 1.6 %, and the value of  $I_w$  is slightly smaller than  $I_u$ . For the linear velocity profiles, the  $I_u$  and  $I_w$  are approximately 7.2 % and 5.7 %, the  $I_u$  is larger than  $I_w$  obviously. For the parabolic velocity profiles, when  $U < 6$  m/s, the turbulence intensities fluctuate at 12.2 % ( $I_u$ ) and 9.3 % ( $I_w$ ); however, when  $U$  greater than 6 m/s, the value of them fluctuates at 10.8 % ( $I_u$ ) and 8.4 % ( $I_w$ ). This interesting phenomenon may be due to the variable trend along with the parabolic profile, the middle position of the parabolic profile may be affected by other wind at the symmetrical location when the wind velocity is higher than one certain value. These high turbulence intensities generated by non-uniform flows may be due to the nonuniformity and instability of the incoming flow along the span, the turbulence intensity can reach 25 % in some field measurements [23].

Three different wind velocity profiles were used to investigate the vibration characteristics of the bridge model with and without railings and median dividers. Based on the wind profiles and whether the railings were installed, six cases were designed, as summarized in Table 2, where the cases without railings and median dividers include C0, C1, and C2; the cases with railings are denoted CR0, CR1, and CR2.

### 3. Results and discussions

#### 3.1. Vibration characteristics

The vertical VIV responses of the taut-strip bridge without and with railings vary with reduced velocity ( $Ur = U/Hf$ ,  $f = 3.4$  Hz) are shown in Fig. 7(a) and (b) respectively. The vertical VIV responses are mainly observed in engineering [16–18,15]. For the bridge without railings (erection stage), the 1st, 2nd, and 3rd structural mode vibrations appear successively with the increase in the reduced velocity, as indicated in Fig. 7(a). For C0, the maximum VIV responses occur at the reduced velocity  $Ur = 3.5$  (the 1st mode),  $Ur = 7.3$  (the 2nd mode), and  $Ur = 11.8$  (the 3rd mode). For C1, the amplitudes of the 1st to 3rd modal VIV decrease by 42 %, 80 %, and 81 % compared with C0 as listed in Table 3. Therefore, an obvious reduction in the vertical VIV response is observed under a linear wind velocity profile. For C2, the 2nd and 3rd modal VIV responses are obtained, while the 1st mode is not observed owing to the minimum flow-velocity limitation of the wind tunnel. The amplitudes of the 2nd and 3rd modal VIVs in C2 decrease by 68 % and 72 %, respectively.



**Fig. 5.** Velocity and turbulence intensity profiles: (a) linear and (b) parabolic velocity profiles under different incoming wind speeds; (c) linear and (d) parabolic turbulence intensity profiles under different incoming wind speeds.

respectively, which is similar to the results for C1. Inevitably, except for the variable wind profiles, the turbulence effects should be considered due to the higher value of the turbulence intensities of the two non-uniform flows. The variable velocity of the surrounding flow field is detrimental to accumulating a stable VIV response [31].

The vibration responses of the taut-strip bridge with railings (completed bridge stage) are analogous to those without railings as illustrated in Fig. 7(b). For CR0, the maximum VIV responses of each modal VIV occur at  $Ur = 10.1$  (the 1st mode),  $Ur = 18.6$  (the 2nd mode), and  $Ur = 25.2$  (the 3rd mode). In addition, the amplitude of the 2nd and 3rd modal VIVs are decreased as well compared with the 1st modal VIV. According to the results in Fig. 7, the lock-in ranges of each modal VIV are considerably wider for CR0 than for C0, which indicates that the VIV responses of the taut-strip model with railings are more apparent than those without railings under a uniform flow. The difference in the VIV amplitudes between the decks with and without railings occurs due to the flow field change, and the different flow separation mainly. The separation point moves upstream and leads to larger leading flow separation as well as a wider wake for the bridge with railings [24,15,25].

For CR1, the values of the dimensionless amplitudes ( $y/H$ ) of the first three modal VIVs exhibit reductions of 17 %, 19 %, and 22 %, respectively, compared with those for CR0. A similar phenomenon was reported by Wang and Xiao [29], where the maximum amplitudes under a linear shear flow were smaller than those under a uniform flow. Additionally, for CR2, the values of  $y/H$  of the three VIV modes exhibit reductions of 25 %, 32 %, and 33 %, respectively, compared with CR0, and these three reductions are larger than those for CR1. The lock-in regions under the two non-uniform profiles are similar to CR0 at the 1st mode and are slightly smaller than CR0 at the 2nd and 3rd modes. Similar results were presented by Jiang et al. [14], who indicated that the lock-in region of a riser under a shear cross flow was narrower than that under a uniform flow. Interestingly, the maximum amplitude reduction of CR1 is less than that of CR2, whereas the reduction of C1 is larger than that of C2 as listed in Table 3. The possible reason is that the installed railings

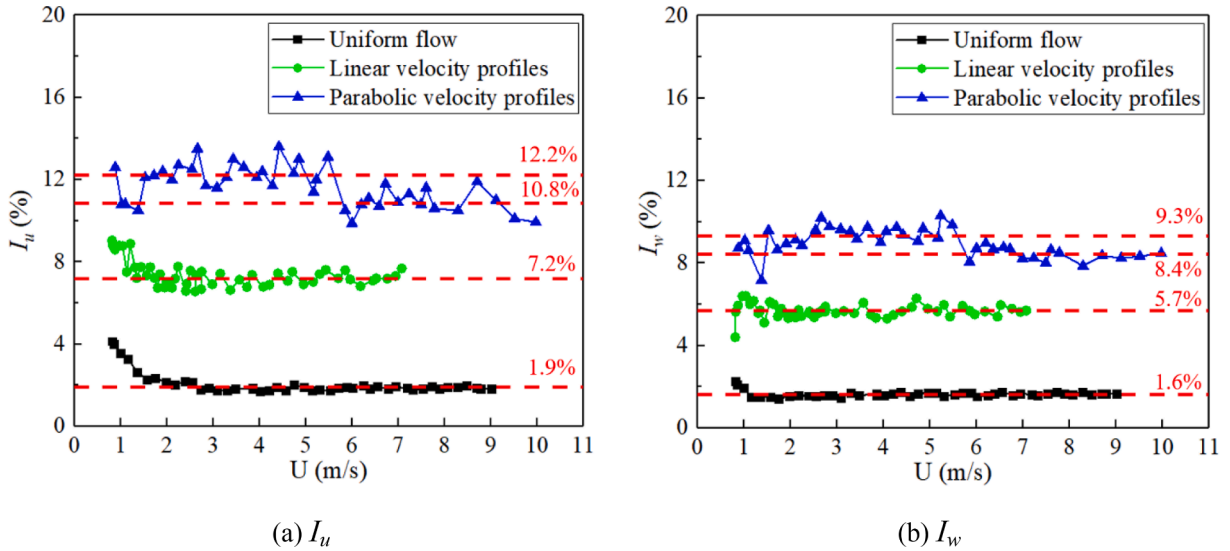


Fig. 6. Turbulence intensities of three types of incoming flows vary with wind velocities.

Table 2

Six test cases.

| Cases            | Uniform flow | Linear wind velocity profile | Parabolic wind velocity profile |
|------------------|--------------|------------------------------|---------------------------------|
| Without railings | C0           | C1                           | C2                              |
| With railings    | CR0          | CR1                          | CR2                             |

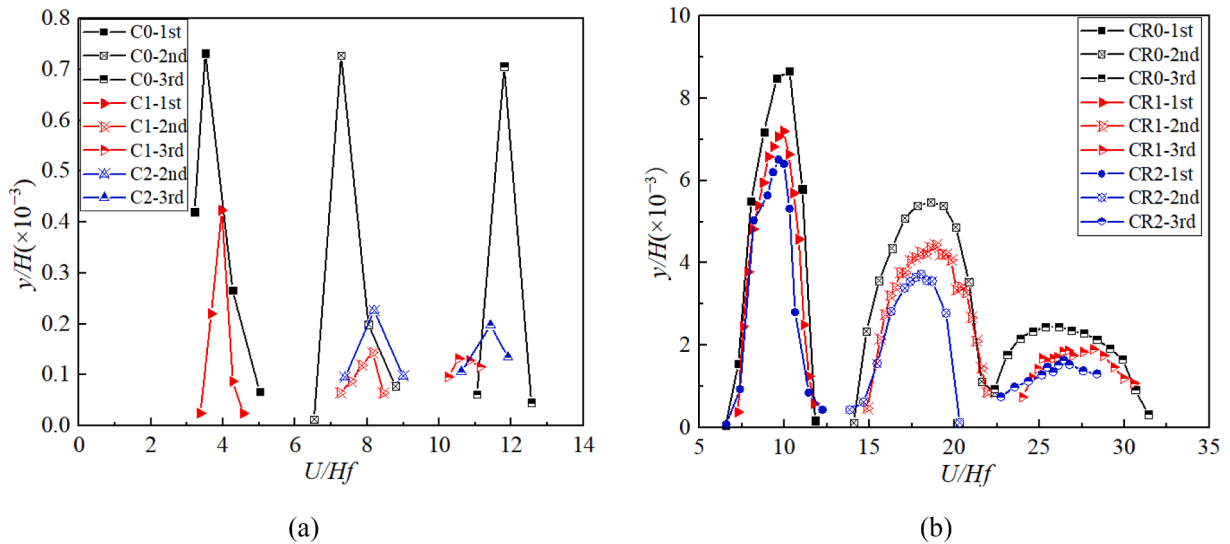


Fig. 7. Amplitudes of displacement vary with reduced velocities: (a) C0, C1, and C2; (b) CR0, CR1, and CR2.

have a promoting effect on the turbulence intensities and wind velocities at the leading edge of a bridge girder [1], the higher turbulence intensities of CR2 lead to the higher reduction of the  $y/H$ .

### 3.1.1. Cases under a uniform flow

To determine the VIV characteristics of each mode and understand the vibration mechanism of multi-mode VIVs, the maximum amplitudes of each modal VIV were extracted to investigate the responses based on time-history and frequency spectrum analyses.

For the uniform flow, the time history and frequency spectrum of each VIV response of the bridge without and with railings are

**Table 3**

Reduction extents of VIV amplitudes of non-uniform flows compared to uniform flows.

| Cases | The 1st mode | The 2nd mode | The 3rd mode |
|-------|--------------|--------------|--------------|
| C1    | 42 %         | 80 %         | 81 %         |
| C2    | –            | 68 %         | 72 %         |
| CR1   | 17 %         | 19 %         | 22 %         |
| CR2   | 25 %         | 32 %         | 33 %         |

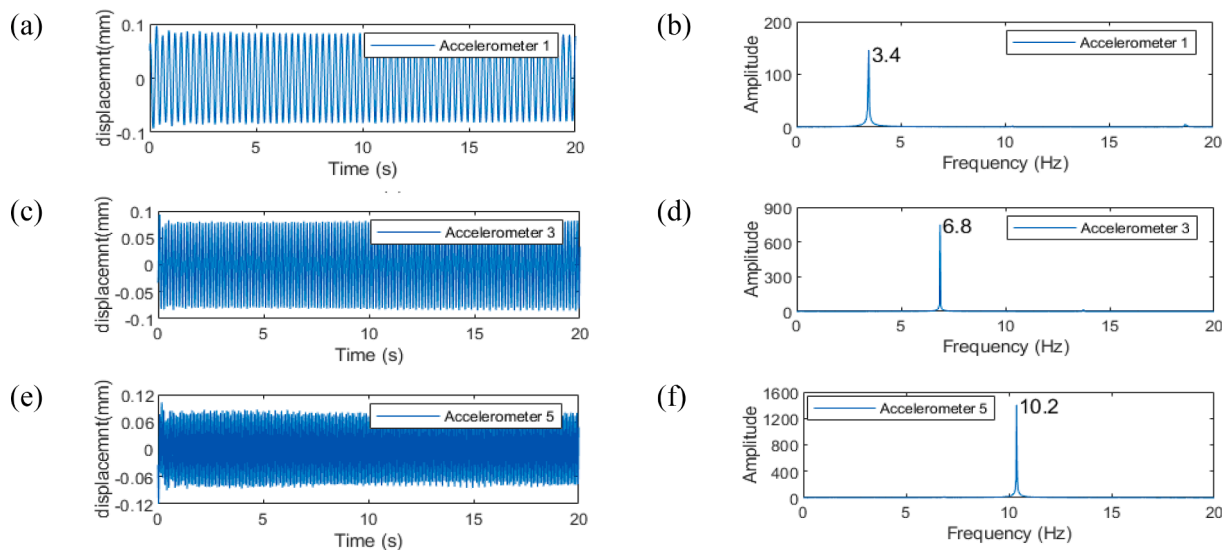
illustrated in Figs. 8 and 9, respectively. For C0, as shown in Fig. 8(a), (c), and (e), the vibration amplitudes are 0.08 mm, 0.08 mm, and 0.079 mm respectively which are consistent with Fig. 7(a), and these amplitudes show a smooth trend. Fig. 8(b), (d), and (f) show that the modal frequencies for the 1st through 3rd modes are 3.4 Hz, 6.8 Hz, and 10.2 Hz, respectively. Only one frequency is present in the spectrum; therefore, a single-mode vibration (from the 1st to 3rd modes) is observed in C0. Vandiver et al. [28] noted that the response spectrum of a single-mode vibration contains only one dominant resonant mode peak, and the secondary peak should be <5 % of the maximum peak.

The CR0 case for the bridge with railings and median dividers is illustrated in Fig. 9. The vibration amplitudes of the first three modes are steady, similar to C0. The modal vibration amplitudes are reduced with an increase in the structural mode frequencies apparently. In addition, the 2nd- to 3rd-order modal frequencies are distinct single peaks in the frequency spectrum. However, the phenomenon of multiple frequencies is observed in Fig. 9(b), with a dominant frequency of 3.4 Hz and a double and a triple frequency of 6.8 Hz and 10.2 Hz in the spectrum. These small amplitudes frequency components are the integer multiples of the predominant frequency.

### 3.1.2. Cases under a linear wind velocity profile

For the linear wind velocity profile, in C1, not only the 1st-order modal frequency but also a higher-order modal frequency (the 5th-order modal frequency) are observed with steady amplitudes, as shown in Fig. 10(a) and (b). The non-uniform and quasi-stable flow with high turbulence intensity brings complex vibration and promotes the generation of high-order harmonic components. Zhang et al. [32] treated the combination of the uniform flow and shear oscillatory flow as the reason for the high-order frequency components of VIV, which leads to extra fatigue damage of flexible slender structures. Two obvious vibration frequencies (3.4 Hz and 6.8 Hz) and three frequencies (3.4 Hz, 6.8 Hz, and 10.2 Hz) are observed at  $Ur = 8.2$  and  $10.5$ , respectively, as shown in Fig. 10(d) and (f). It is worth mentioning that the predominant frequency at  $Ur = 8.2$  is the 2nd-order modal frequency, while the predominant frequency at  $Ur = 10.5$  is the 3rd modal frequency. With an increase in  $Ur$ , many higher-order modal frequency components appear in the frequency spectra. For  $Ur = 15.6$ , more than five frequency components appear, and the 4th-order mode is predominant, as shown in Fig. 10(h). As shown in Fig. 10(c), (e), and (g), these multiple frequencies compete with each other with unsmooth vibration amplitudes during the sampling time, resulting in complicated frequency components and reduction of displacement amplitudes. This phenomenon explains the significant amplitudes reduction of the C1 as mentioned in Fig. 7(a) and demonstrates that multiple frequencies disturb the regular vortex shedding and decrease the VIV response, as reported by Zhu et al. [38].

For CR1, Fig. 11(a), (c), and (e) show steady vibration amplitudes at the 1st and 2nd modes and an unsteady vibration amplitude in the time history at the 3rd mode. In addition, a decreasing trend in the vibration amplitudes is observed mode by mode. The frequency



**Fig. 8.** Vibration responses of C0 at different reduced velocities: (a), (b)  $Ur = 3.51$ ; (c), (d)  $Ur = 7.28$ ; (e), (f)  $Ur = 11.80$ .

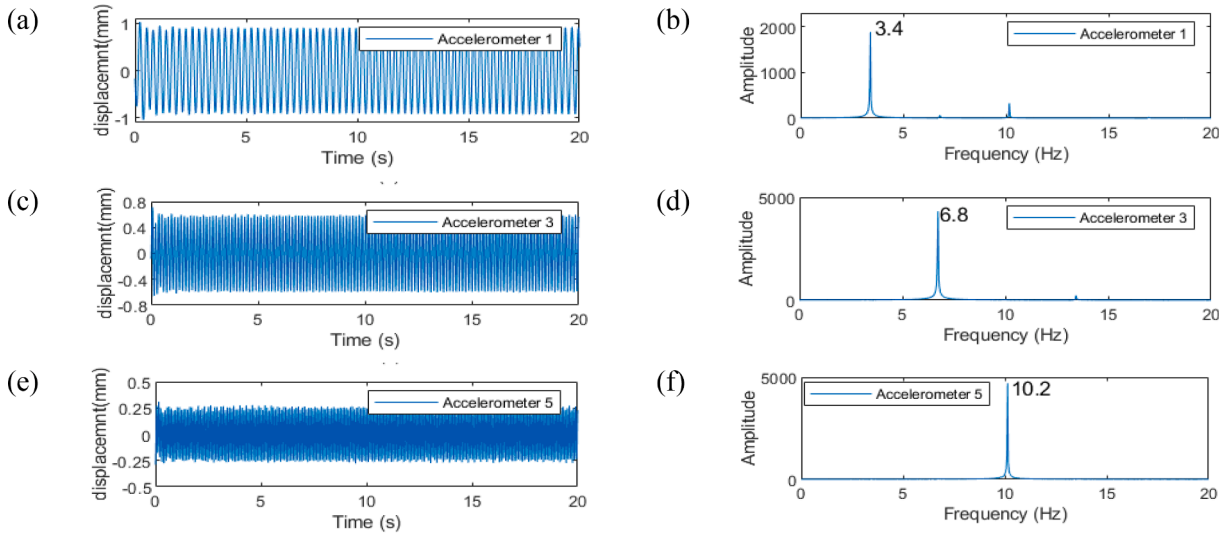


Fig. 9. Vibration responses of CR0 at different reduced velocities: (a), (b)  $Ur = 10.1$ ; (c), (d)  $Ur = 18.6$ ; (e), (f)  $Ur = 25.2$ .

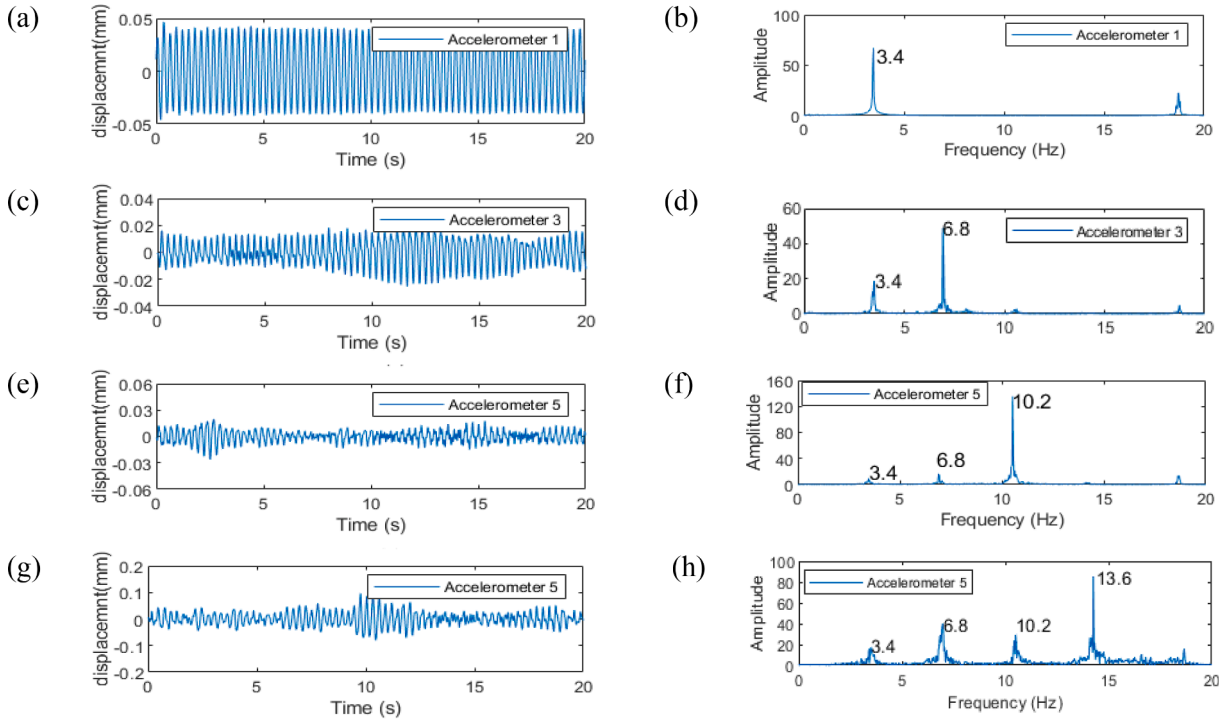


Fig. 10. Vibration responses of C1 at different reduced velocities: (a), (b)  $Ur = 4.0$ ; (c), (d)  $Ur = 8.2$ ; (e), (f)  $Ur = 10.5$ ; (g), (h)  $Ur = 15.6$ .

spectrum of the 1st modal vibration exhibits a 1st-order modal frequency with a large amplitude, while a small amplitude is recorded for a frequency of 10.2 Hz. Similar to CR0, this small amplitude of frequency is triple the dominant frequency (3.4 Hz). For  $Ur = 18.9$  and 26.7, however, each frequency spectrum exhibits only one peak, as shown in Fig. 11(d) and (f), respectively, which indicates that these two vibrations are single-mode vibrations. For  $Ur = 33.5$ , more modal frequency components appear at high  $Ur$ .

### 3.1.3. Cases under a symmetrical parabolic wind velocity profile

For the symmetrical parabolic wind velocity profile, the time history and frequency spectra of the vibrations for C2 are presented in Fig. 12. For  $Ur = 8.2$ , the 2nd-order modal frequency is predominant in the frequency spectrum, and the 1st-order modal frequency component appears with a small proportion. For  $Ur = 11.4$ , the predominant frequency is 10.2 Hz, while peaks are also observed at 3.4



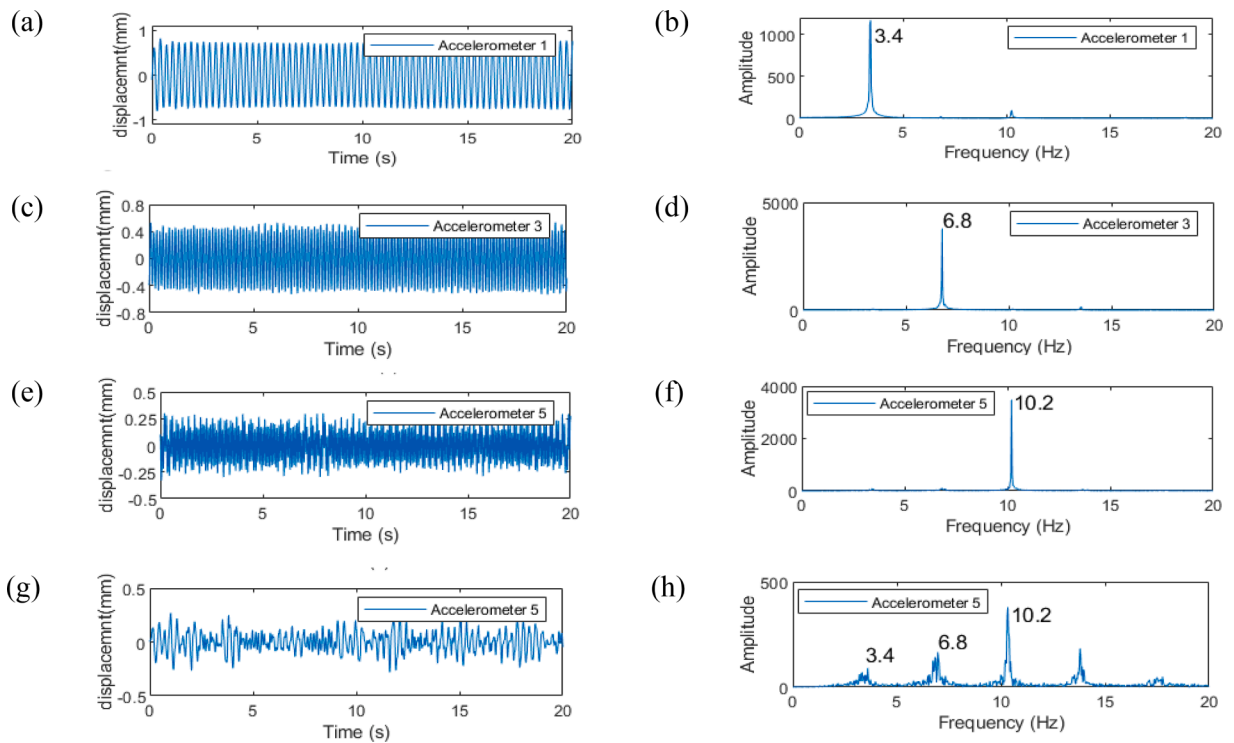


Fig. 11. Vibration responses of CR1 at different reduced velocities: (a), (b)  $Ur = 9.9$ ; (c), (d)  $Ur = 18.9$ ; (e), (f)  $Ur = 26.7$ ; (g), (h)  $Ur = 33.5$ .

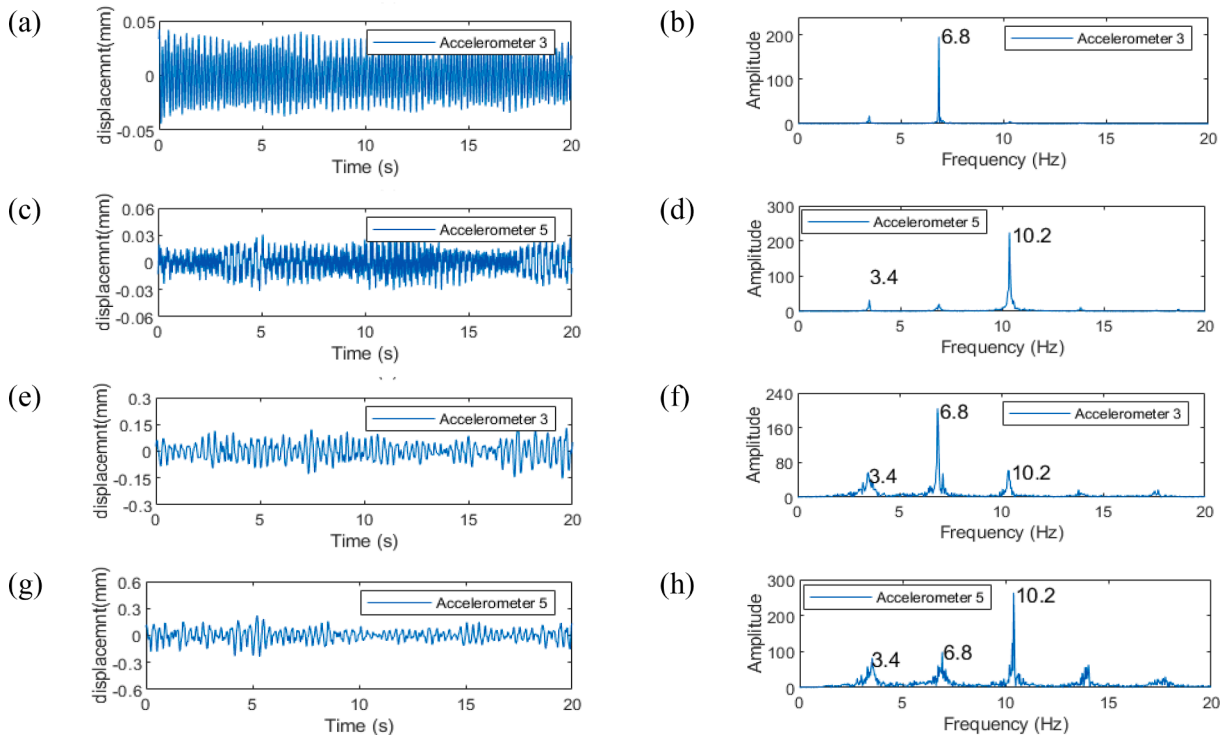


Fig. 12. Vibration responses of C2 at different reduced velocities: (a), (b)  $Ur = 8.2$ ; (c), (d)  $Ur = 11.4$ ; (e), (f)  $Ur = 17.1$ ; (g), (h)  $Ur = 25.2$ .

Hz and 6.8 Hz. For  $Ur = 17.1$  and  $25.2$ , the predominant modal frequencies are the 2nd and 3rd modal frequencies, respectively, and the frequency spectra include other modal frequency components, leading to an unsteady vibration amplitude in the time history, as shown in Fig. 12(e) and (g). Similar to C1 where the multi-frequencies lead to unsteady and lower amplitudes during multi-mode VIVs, the amplitude decrease of C2 is smaller due to the different wind velocity profiles. The displacement ( $y$ ) of the 2nd mode of C2 is  $< 0.05$  mm and  $y$  approaches 0.02 mm for C1. The higher the shear parameter of the wind profile, the more competition would occur among the multiple modes, and the lower the amplitudes of the risers would be [19].

In the case of CR2, for  $Ur = 9.6$  and  $18.4$ , the vibration amplitudes are steady and the frequency spectra show a single dominant frequency in Fig. 13(a)–(d), illustrating that single-mode vibrations occur. At  $Ur = 26.4$ , the dominant frequency is the 3rd modal frequency of 10.2 Hz, while components of the 1st and 2nd modal frequency components are also observed. The time history becomes unsteady, in contrast to the results at  $Ur = 9.6$  and  $18.4$ .

### 3.2. Wake characteristics

The vortex-shedding frequencies at middle span in the wake of the taut-strip model were normalized using the 1st-order modal natural frequency. In this section, the normalized vortex-shedding frequencies ( $f/f_0$ ,  $f_0 = 3.4$  Hz) shown vary with the reduced velocity in Fig. 14 for the six cases. The  $St$  of the bridge without railings is approximately 0.25, whereas that for the bridge with railings is only 0.12. Initially, the normalized vortex-shedding frequencies increase with an increase in the reduced velocity. When  $f/f_0$  approaches 1, the wake frequency is close to the 1st-order modal frequency, and the wake frequency is locked at a certain velocity range. Then, the wake frequency increases again, following the trend in the  $St$  values. For the bridge without railings (Fig. 14(a)–(c)), C0 has more obvious lock-in regions than C1 and C2 (similar to Fig. 7). These weakened lock-in regions indicate that multi-mode frequencies disturb the lock-in process, as reported by Vandiver et al. [28]. For the bridge with railings, the lock-in regions are wider, and the multiple-frequency phenomenon is less obvious than for the bridge without railings.

#### 3.2.1. Wake characteristics of the bridge without railings

The results for the bridge without railings under uniform and non-uniform flows are illustrated in Fig. 15. Fig. 15(a) shows the wake frequencies under a uniform flow along the span, where each wake frequency at the measurement point is locked at the corresponding modal frequency. When the incoming flow is a non-uniform flow, as for C1, four velocities are selected to measure the wake frequency, as shown in Fig. 15(b). When  $f/f_0$  approaches 1, the wake frequency is locked at the 1st-order modal frequency, and it is difficult to excite other modal frequencies. When  $f/f_0$  is larger than 1, the tendency of the normalized vortex-shedding frequencies for C1 is similar to the trend for the linear wind velocity profile. When the wind speed increases, the difference between the two ends of the linear profile also increases. Over two structural frequencies are excited simultaneously at different locations along the span, the vibration

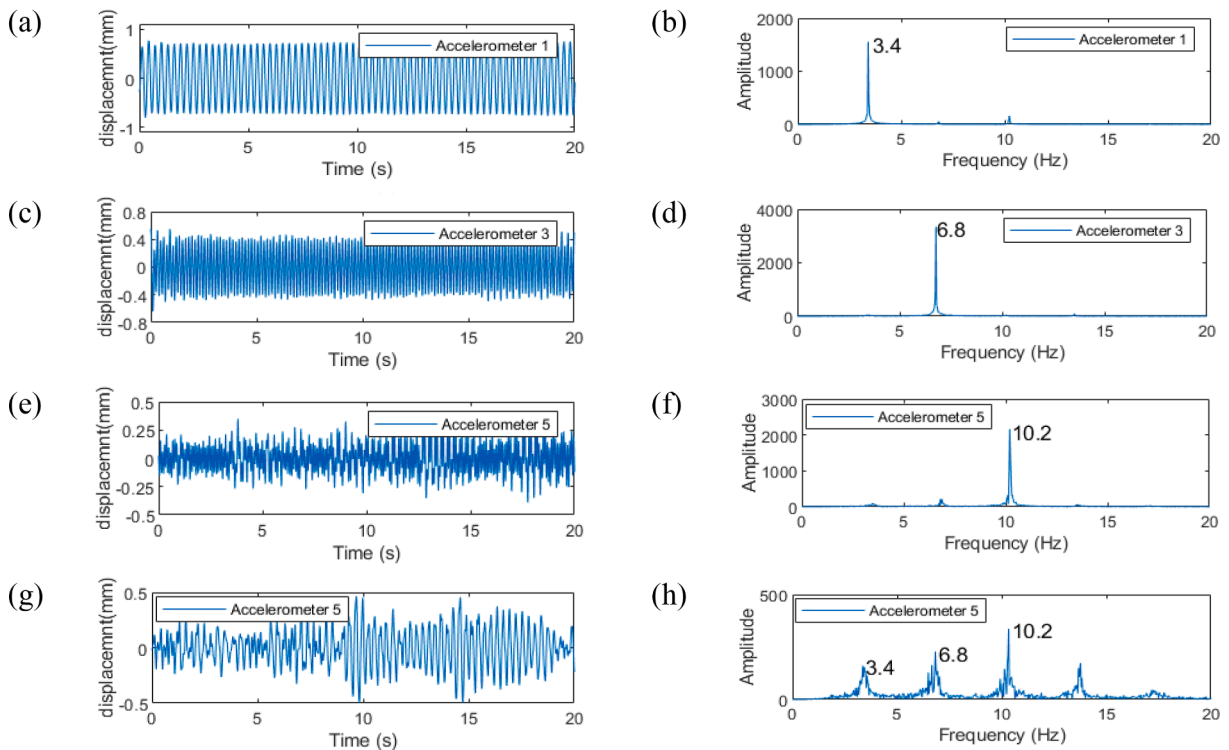


Fig. 13. Vibration responses of CR2 at different reduced velocities: (a), (b)  $Ur = 9.6$ ; (c), (d)  $Ur = 18.4$ ; (e), (f)  $Ur = 26.4$ ; (g), (h)  $Ur = 31.6$ .

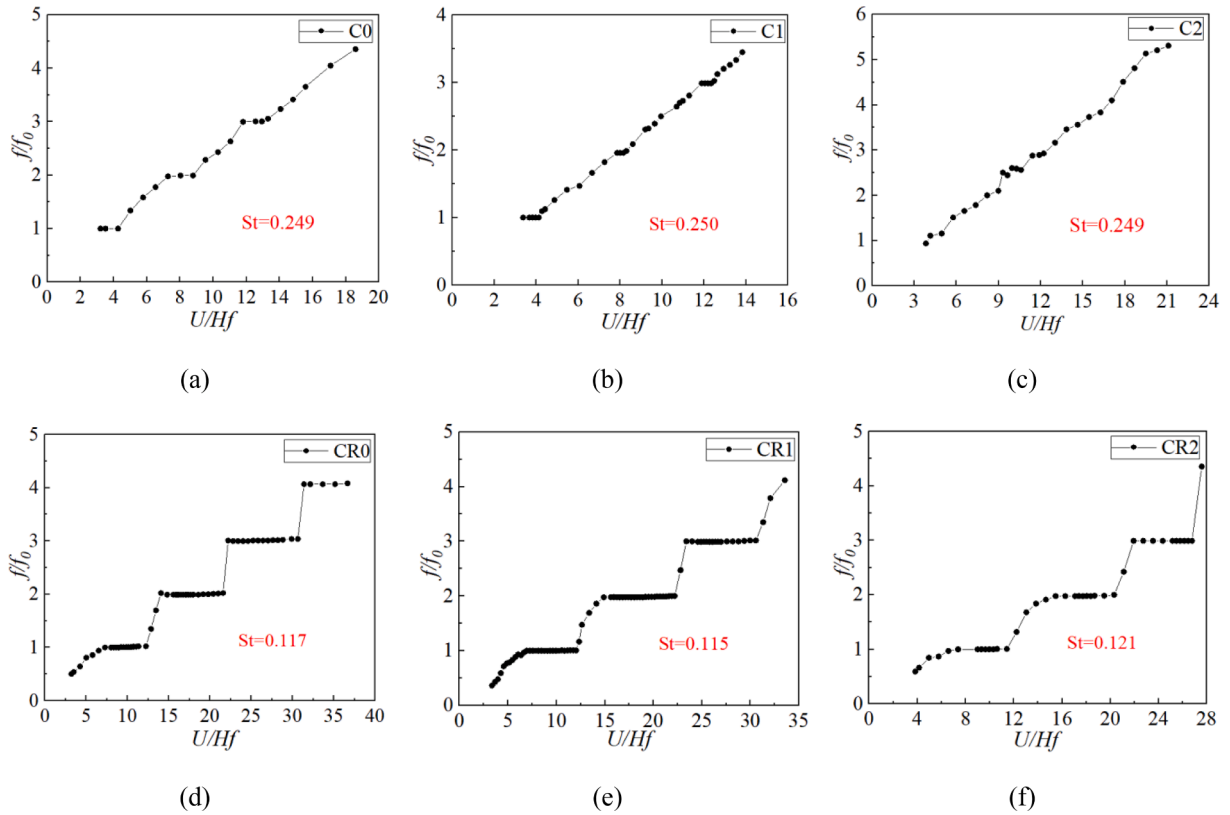


Fig. 14. Normalized vortex-shedding frequencies vary with reduced velocities: (a) C0; (b) C1; (c) C2; (d) CR0; (e) CR1; (f) CR2.

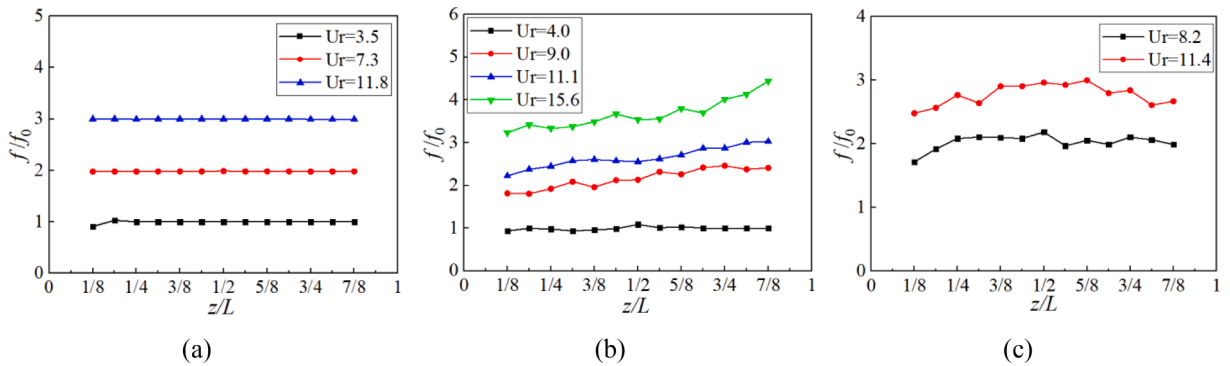
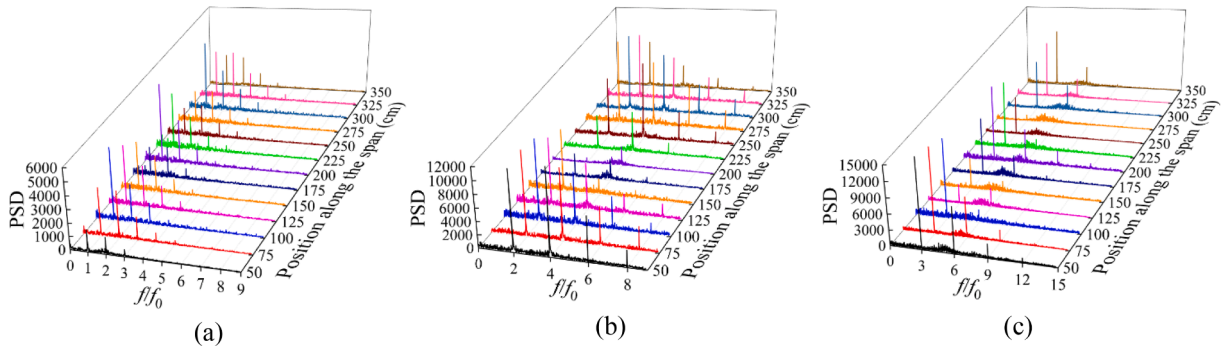


Fig. 15. Normalized vortex-shedding frequencies along the span of the taut-strip model without railings under different wind speeds: (a) C0; (b) C1; (c) C2.

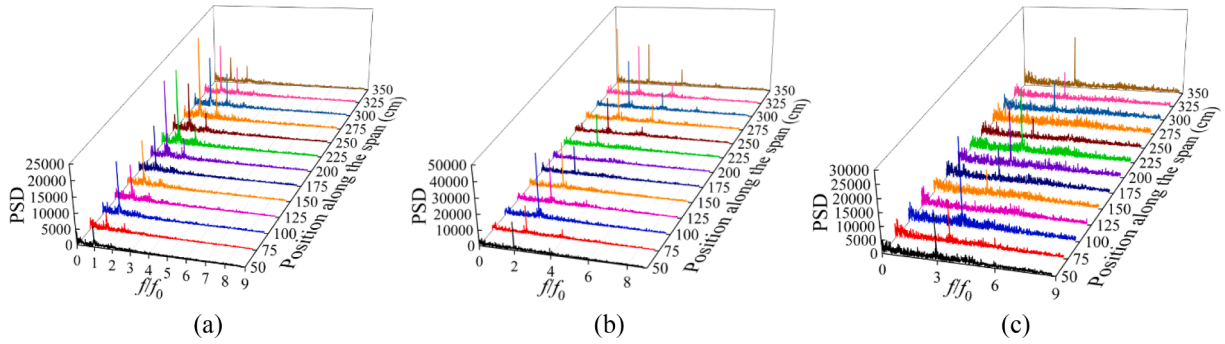
extends to the whole span with several modal frequencies. However, the wake frequencies vary with the linear wind profile and not be locked to one modal frequency due to the weak VIV response. The multi-mode VIV responses (C1 and C2) are much smaller than single-mode VIV responses (C0) refer to Fig. 7(a). The dominant participating modal frequency varies along the length of the flexible structure [38], and the wake frequencies vary along the structural span due to the span-wise coupling of vortices [19]. For C2 (Fig. 15(c)), the frequencies increase until  $z/L$  approaches  $1/2$ , when the frequencies begin to decrease. The shape thus resembles a parabolic curve, similar to the situation under the parabolic wind velocity profile.

### 3.2.2. Wake characteristics of the bridge with railings

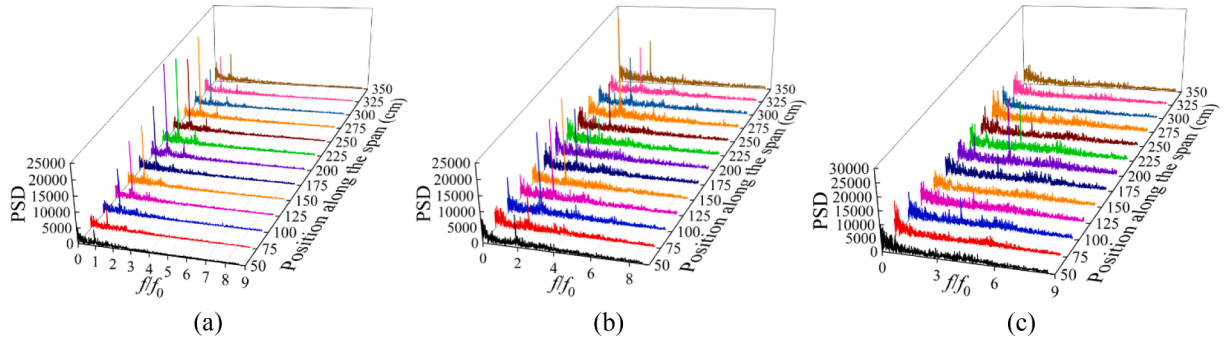
For the bridge with railings and median dividers, the variations in the normalized vortex-shedding frequencies of each measurement point along the span are shown in Figs. 16–18, subplots (a)–(c) represent the 1st–3rd modal vibration. The multiple-frequency phenomenon of the bridge with railings appears in the frequency spectra. For a uniform incoming flow with  $Ur = 10.1$ , the



**Fig. 16.** Normalized vortex-shedding frequencies along the span of the taut-strip model (CR0) under different wind speeds: (a)  $Ur = 10.1$ ; (b)  $Ur = 18.6$ ; (c)  $Ur = 25.2$ .

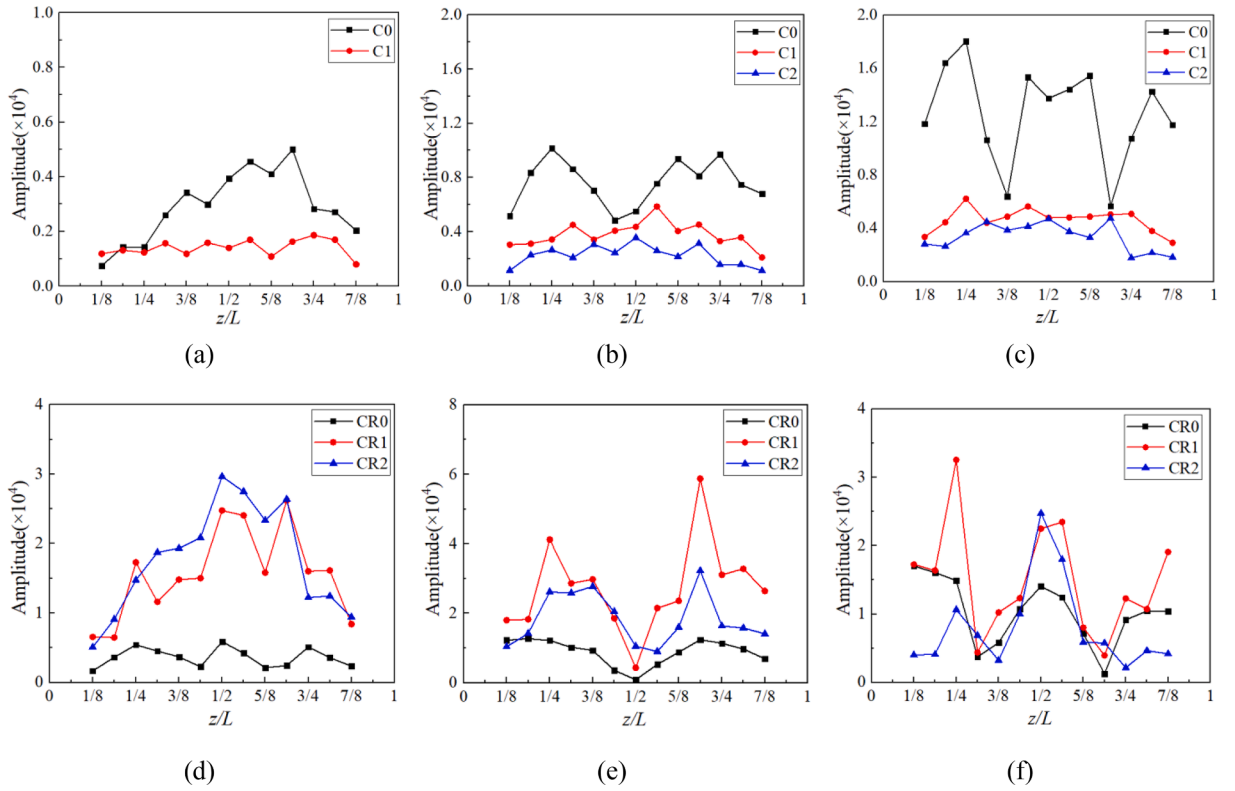


**Fig. 17.** Normalized vortex-shedding frequencies along the span of the taut-strip model (CR1) under different wind speeds: (a)  $Ur = 9.9$ ; (b)  $Ur = 18.9$ ; (c)  $Ur = 26.7$ .

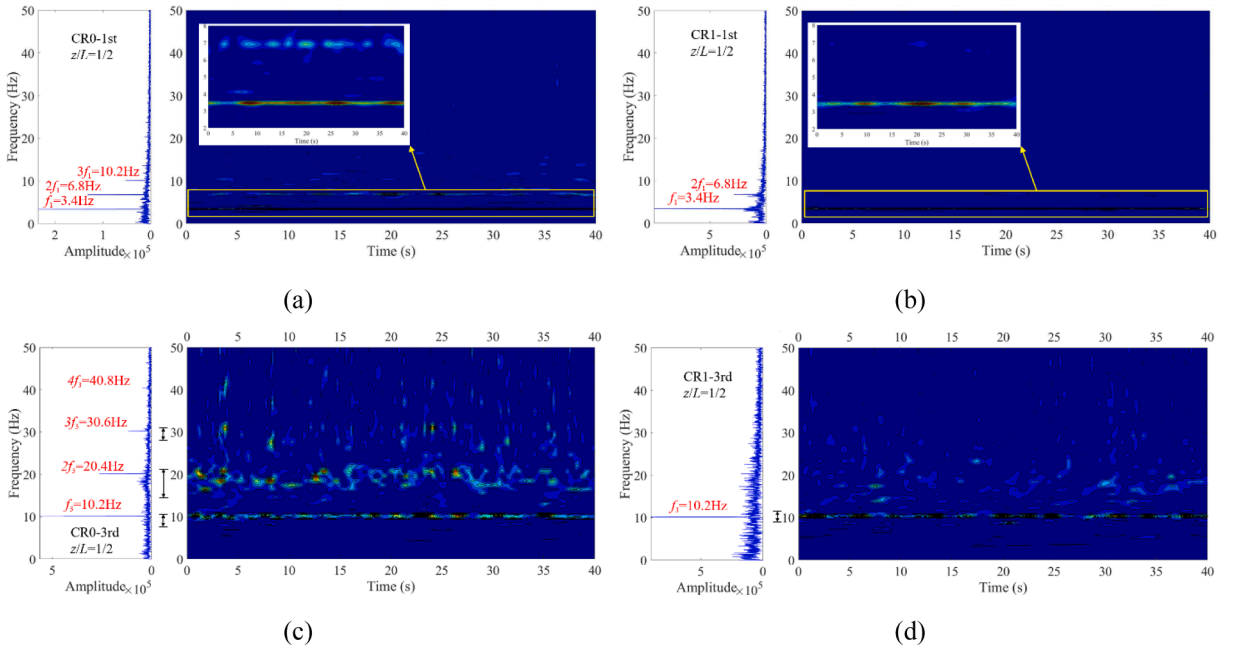


**Fig. 18.** Normalized vortex-shedding frequencies along the span of the taut-strip model (CR2) under different wind speeds: (a)  $Ur = 9.6$ ; (b)  $Ur = 18.4$ ; (c)  $Ur = 26.4$ .

predominant frequency is 3.4 Hz, i.e.,  $f/f_0 = 1$ ; however, normalized frequencies  $f/f_0 = 2, 3$ , and 4 appear in the spectrum, which are the integer multiples of the predominant frequency. The integer-multiple frequency phenomenon also appears in the 2nd and 3rd modal vibrations, which indicates that the VIV response is intense under uniform flow. In addition, the frequencies vary along the span, and the amplitudes of the predominant frequencies are illustrated in Fig. 19. For the non-uniform incoming flow, as shown in Figs. 16 and 17, the integer-multiple frequency phenomenon is less obvious than for the bridge under uniform flow. Because the non-uniform flows and higher turbulence intensities disturb the regular vortices in the wake, the integer-multiple frequencies decrease markedly, and are even insignificant, further indicating that non-uniform flows suppress the VIV responses. Besides, the measured structural vibration is a coupled vibration compared with each local vibration, and the vortex-shedding frequencies can show the local vortex shedding at any position along the span. The vortex-shedding frequencies at different locations are different, for example in Fig. 15(b)  $z = 250$  and 300 cm, the maximum  $f/f_0$  is 4 and 2.



**Fig. 19.** Amplitudes of wake frequencies along the span of the taut-strip model: (a)–(c) the 1st–3rd modes of the bridge without railings; (d)–(f) the 1st–3rd modes of the bridge with railings.



**Fig. 20.** Time-frequency responses of the midspan of wake frequencies along the span: (a) the 1st mode of CR0; (b) the 1st mode of CR1; (c) the 3rd mode of CR0; (d) the 3rd mode of CR1.

### 3.2.3. Wake characteristics along the span

The amplitudes of the predominant frequencies in the wake are selected along the span to further explore the frequency characteristics, as shown in Fig. 19. For the uniform flow (black points in Fig. 19(a)–(f)), the maximum amplitudes of the 2nd- and 3rd-order modes are almost twice and four times that of the 1st-order respectively, indicating that the higher-order modes are prone to have stronger energy generated from wake vortices with a multiplying trend [20]. Furthermore, the amplitudes of each mode show obvious mode shape characteristics, illustrating that the wake vortices along the span are affected by the corresponding mode shape. For a flexible structure, the amplitudes present a mode shape trend [36].

For the bridge without railings under non-uniform flows, as shown in Fig. 19(a)–(c), the amplitude distribution of each mode has little change and smaller value than those under uniform flow. The turbulence intensities generated from the non-uniform and the mode competition between multi-frequencies consume vortex energy and suppress the response. Thus, the bridge in the erection stage is relatively safe under non-uniform flow. For the bridge with railings under non-uniform flows, as shown in Fig. 19(d)–(f), the predominant frequency amplitudes for CR1 and CR2 are higher than those for CR0 because of the obvious integer-multiple frequency components of CR0 (see Figs. 15–17). Besides, the amplitudes of CR1 and CR2 follow a trend of mode shape similar to CR0, this may due to the railings which amplify the flow separation at the leading edge of the bridge and lead to stronger VIV responses than the one without railings (C1 and C2).

The time-frequency responses of the midspan ( $z/L = 1/2$ ) of CR0 and CR1 are given in Fig. 20. The maximum responses of the 1st and 3rd modes both occurred at the midspan of the bridge. For the 1st mode of CR0, the dominant frequency ( $f_1$ ) is 3.4 Hz. A double ( $2f_1$ ) and a triple ( $3f_1$ ) frequencies are observed as well in Fig. 20(a). For CR1 (Fig. 20(b)), the integer-multiple frequencies and high-order frequencies are not obvious as in CR0. The non-uniform flow interferes with the regular shedding of vortices and suppresses the VIV responses. For the 3rd mode of CR0, the dominant frequency ( $f_3$ ) is 10.2 Hz, the shedding of vortices is time-varying, and more integer-multiple frequency components existed in Fig. 20(c). The majority of the vortex energy is concentrated in the dominant frequency, and many discrete small vortices follow the bridge. For CR1, the phenomenon is less obvious than in CR0 and the higher-order frequency components are irregular due to the non-uniformness of the incoming flow. Due to the increment of wind velocity and structural mode (from the 1st to 3rd modes), the frequency bandwidth turns broader and some higher-order frequency components are excited correspondingly both under uniform and non-uniform flows. The shedding of vortices is intermittent due to the instability of VIV at high velocities [32].

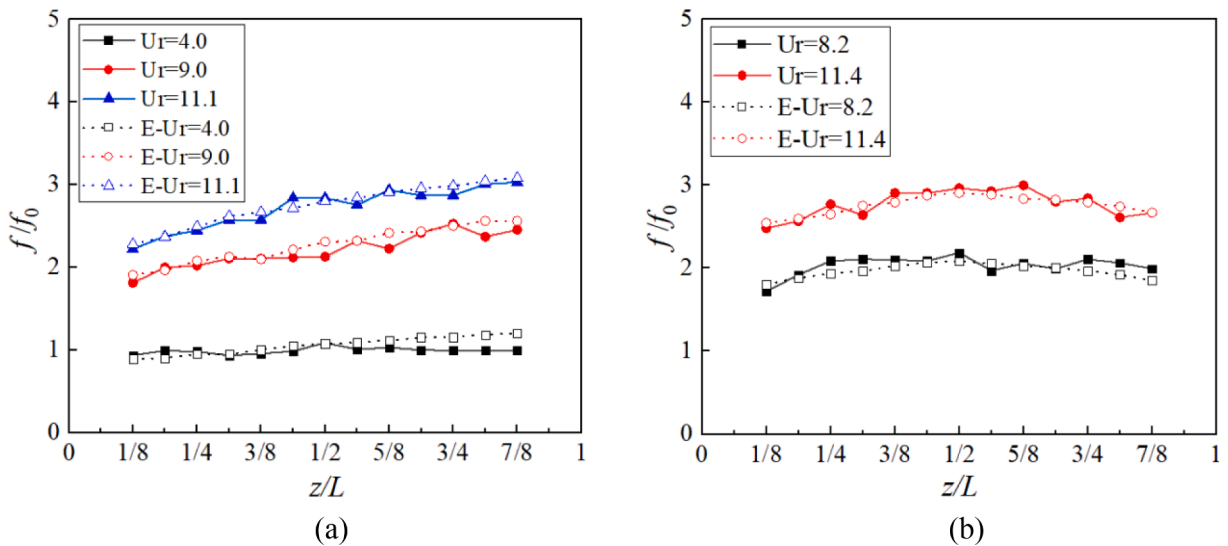
### 3.2.4. Comparisons of estimated and measured results

The vortex-shedding frequency of a flexible bridge model can be expressed as follows:

$$f(z) = \text{St} \frac{U_0(z)}{H} \quad (1)$$

where  $f(z)$  is the estimated vortex-shedding frequency along the span with the length of  $z$ ,  $U_0(z)$  is the incoming wind velocity along the span with the length of  $z$ , and  $H$  is the height of the taut-strip model; for the bridge without railings,  $\text{St} = 0.25$ .

Referring to Eq. (1), the estimated vortex-shedding frequency along the span direction can be predicted as illustrated in Fig. 21, where the solid and dashed lines represent the measured and estimated wake frequencies, respectively. For C1 with  $Ur = 4$ , the estimated wake frequencies increase along the span, while the measured wake frequencies are locked at the 1st modal frequency, as



**Fig. 21.** Measured and estimated vortex-shedding frequencies along the span of the taut-strip model without railings under different wind speeds: (a) C1; (b) C2.



shown in Fig. 21(a). For  $Ur = 9.0$  and  $11.1$ , the estimated wake frequencies are close to the measured wake frequencies. Similarly, for C2, the estimated vortex-shedding frequencies match the measured wake frequencies, as shown in Fig. 21(b). If the incoming wind velocity along the span is known, the wake frequencies of the flexible bridge without railings can be predicted initially using Eq. (1).

### 3.3. Identification of multi-mode VIV

The estimated vortex-shedding frequencies calculated by Eq. (1) under a linear wind profile at  $Ur = 10.4$  of C1 are shown in Fig. 22. The multi-mode VIVs occur and the wake frequencies along the span cover the 2nd and 3rd modal frequencies simultaneously. Therefore, when the wind difference is high enough to excite two or more modes along the span, the VIV of the bridge without railings is no longer a single mode, and several modes exist at the same time at different places along the span. Due to the non-uniform flows, the local vibration of the bridge sections at different lengths of the span is different, the local vibration at one place may spread to another place towards one or both ends along the span [10,2]. The local vibration of the bridge affects by the shedding of vortices as well as the vibration of the adjacent or farther sections of the bridge. Thus, the multi-mode VIV becomes an intricate vibration due to the multiple frequencies consisting simultaneously along the bridge span.

The incoming flow of a linear wind profile expressions as.

$$U(z) = a + bz \quad (2)$$

where  $a$  and  $b$  are the parameters of intercept and slope of the linear wind profile, the minimum and maximum wind velocity are given by.

$$U_{\min} = \begin{cases} U(0) & b > 0 \\ U(L) & b < 0 \end{cases} \quad (3)$$

$$U_{\max} = \begin{cases} U(L) & b > 0 \\ U(0) & b < 0 \end{cases} \quad (4)$$

Similarly, the incoming flow of a symmetrical parabolic wind profile expressions as.

$$U(z) = a + c\left(z - \frac{L}{2}\right)^2 \quad (5)$$

where  $c$  determines the parabola going upwards or downwards, the minimum and maximum wind velocity are given by.

$$U_{\min} = \begin{cases} U(L/2) & c > 0 \\ U(0) = U(L) & c < 0 \end{cases} \quad (6)$$

$$U_{\max} = \begin{cases} U(0) = U(L) & c > 0 \\ U(L/2) & c < 0 \end{cases} \quad (7)$$

The corresponding minimum and maximum wake frequencies follow the St law and can be expressed as.

$$f_{\min} = St \frac{U_{\min}}{H} \quad (8)$$

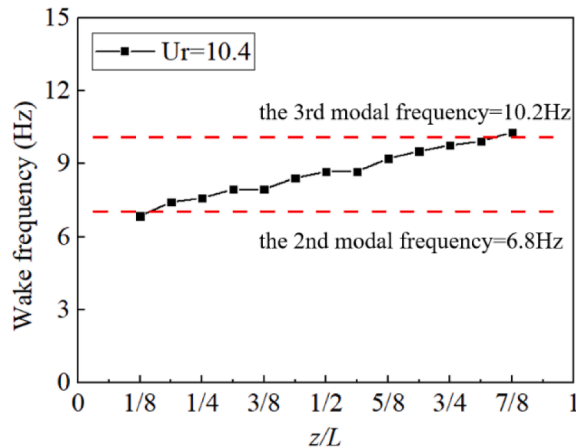


Fig. 22. Estimated wake frequencies versus velocity profile of C1.

$$f_{\max} = \text{St} \frac{U_{\max}}{H} \quad (9)$$

When the multi-mode VIV occurs, the natural frequencies of the bridge of the  $n$ th mode VIV satisfy the following:

$$f_{\min} \leq f_n \leq f_{\max} \text{ (the number of } f_n \geq 2) \quad (10)$$

For example, the incoming flow profile in Fig. 22 is selected to validate this method, the minimum and maximum wake frequencies are 6.6 Hz and 10.3 Hz respectively. Substituting  $f_{\min} = 6.6$  and  $f_{\max} = 10.4$  into Eq. (10), that is,  $f_{\min} < 6.8 < 10.2 < f_{\max}$ . Therefore,  $f_n$  contains  $f_2$  and  $f_3$ , the vibration of the bridge model under this wind profile is multi-mode VIV and the predominant structural mode is the 2nd and 3rd mode.

#### 4. Conclusion

In this study, the characteristics of multi-mode VIVs under non-uniform wind velocity profiles were investigated in the wind tunnel. Based on the vibration responses and wake characteristics of a taut-strip model, the main conclusions are as follows.

1. A non-uniform wind velocity profile disturbs the regular vortices in the wake because of the variable wind velocity and higher turbulence intensity, leading to suppression of the VIV response evidently; the more frequency components are present, the smaller the amplitudes of the vibrations will be.
2. The extensive difference between the bridge with and without railings is due to the differing separation point, leading to larger leading flow separations, wider wake regions, and larger VIV responses when the bridge with railings.
3. The excitation mechanism of the multi-mode VIVs is that the differences between the maximum and minimum of the non-uniform wind velocity profiles can cause a difference threshold in the wake frequencies. If this covers two or more modal frequencies, multi-mode VIVs will occur. The multiple frequencies compete with each other and deplete their energy, and different frequency components distribute at different spanwise locations simultaneously. A prediction approach of the wake frequencies and an identification method of the multi-mode VIV of the flexible bridge without railings are proposed.
4. With the increase of the wind velocity and structural mode, the shedding of vortices is intermittent, and more high-order frequency components and discrete small vortices are excited correspondingly due to the nonlinearity and instability of VIV. For the bridge without railings, the wake frequencies vary with the trend in the corresponding profiles and their amplitudes follow a trend of mode shape; for the bridge with railings, the wake frequencies are locked at the modal frequencies and are affected by mode shape as well, leading to a strong 3D effect.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Further reading

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